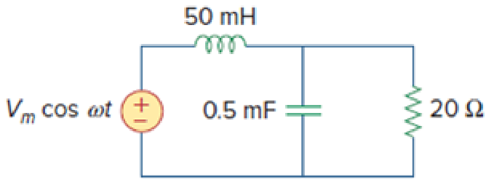
Chapter 14, Problem 9PP

Problem

Calculate the resonant frequency of the circuit in Fig. 14.29.

Figure 14.29:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in Figure 14.29 in the text book.

The inductor in frequency domain is,

$$X_L = j\omega L$$
$$= j\omega\left(50\times10^{-3}\right)$$
$$= 0.05\, j\omega$$

The capacitor in frequency domain is,

$$X_C = \frac{1}{j\omega C}$$
$$= \frac{1}{j\omega\left(0.5\times10^{-3}\right)}$$
$$= \frac{2000}{j\omega}$$

Step 2 of 3

In the circuit, resistor and capacitor are connected in parallel and this parallel combination is in series with inductor.

Calculate the total admittance of the circuit.

$$Y = \frac{\left(\frac{1}{X_C}+\frac{1}{R}\right)\left(\frac{1}{X_L}\right)}{\frac{1}{X_C}+\frac{1}{R}+\frac{1}{X_L}}$$

$$= \frac{\left(\frac{j\omega}{2000}+\frac{1}{20}\right)\left(\frac{1}{0.05j\omega}\right)}{\frac{j\omega}{2000}+\frac{1}{20}+\frac{1}{0.05j\omega}}$$

$$= \frac{\left(\frac{j20\omega+2000}{40000}\right)\left(\frac{1}{0.05j\omega}\right)}{\frac{-\omega^2+j100\omega+40000}{j2000\omega}}$$

$$= \frac{j20\omega+2000}{-\omega^2+j100\omega+40000}$$

$$= \frac{(j20\omega+2000)(40000-\omega^2-j100\omega)}{(40000-\omega^2+j100\omega)(40000-\omega^2-j100\omega)}$$

$$= \frac{j800000\omega-j20\omega^3+2000\omega^2+80\times10^6-2000\omega^2-j200000\omega}{(40000-\omega^2+j100\omega)(40000-\omega^2-j100\omega)}$$

$$= \frac{80\times10^6-j(20\omega^3-600000\omega)}{(40000-\omega^2+j100\omega)(40000-\omega^2-j100\omega)}$$

Step 3 of 3

At resonance imaginary part of the admittance is zero.

$$\mathrm{Im}(Y)=0$$

$$\frac{20\omega^3-600000\omega}{(40000-\omega^2+j100\omega)(40000-\omega^2-j100\omega)}=0$$

$$20\omega^3-600000\omega=0$$

$$20\omega^3=600000\omega$$

$$\omega^2=30000$$

$$\omega=173.21\text{ rad/s}$$

Therefore, the resonant frequency, ω_0 of the circuit is $\boxed{173.21\text{ rad/s}}$.